Main Tank Transformer Oil Standards (Mineral Oil)

Test Item	Standard	Method	Main Tank			
			Std. IEEE C57.106-2015	≤ 69 kV	> 69 kV - < 230 kV	≥ 230 kV
			Std. IEC 60422	≤ 72.5 kV	> 72.5 kV - ≤ 170 kV	> 170 kV - < 400 kV
I. New mineral oil (Received from the oil supplier)						
Dielectric breakdown	IEEE C57.106-2015	ASTM D1816	1 mm. gap (Spherically capped), kV	≥ 20		
			2 mm. gap (Spherically capped), kV	≥ 35		
		ASTM D877	2.54 mm. or 0.1 in. gap (Disks), kV	> 30		
Dissipation factor (Power Factor)	IEEE C57.106-2015	ASTM D924	at 25 °C, % maximum	≤ 0.05		
			at 100 °C, % maximum	≤ 0.3		
Conductivity				Suggested Limit		
Water Content	IEEE C57.106-2015	ASTM D1533	ppm (Mg/kg) maximum	≤ 35		
Color	IEEE C57.106-2015	ASTM D1500	Units maximum	≤ 0.5		
Interfacial tension (IFT)	IEEE C57.106-2015	ASTM D971	mN/m	≥ 40		
Neutralization number (Acidity)	IEEE C57.106-2015	ASTM D974	Mg KOH/g maximum	≤ 0.03		
Oxidation inhibitor content	IEEE C57.106-2015	ASTM D2668	Type I oil, % maximum	≤ 0.08		
			Type II oil, % maximum	≤ 0.3		
Corrosive sulfer	IEEE C57.106-2015	ASTM D1275	-	Non corrosive		
Relative density (Specific Gravit	IEEE C57.106-2015	ASTM D1298	at 15 °C maximum	≤ 0.91		
Visual examination	IEEE C57.106-2015	ASTM D1524	-	Bright and Clear		
II. New mineral oil (Received in new equipment or after filling,prior to energization)						
Dielectric breakdown	IEEE C57.106-2015	ASTM D1816	1 mm. gap (Spherically capped, kV)	≥ 25	≥ 30	≥ 35
			2 mm. gap (Spherically capped, kV)	≥ 45	≥ 55	≥ 60
	IEC 60422	IEC 60156	2.5 mm. gap (Spherical or Spherically capped, kV)	> 55	> 60	> 60
Dissipation factor (Power Factor)	IEEE C57.106-2015	ASTM D924	at 25 °C, % maximum	≤ 0.05		
			at 100 °C, % maximum	≤ 0.4	-	≤ 0.3
	IEC 60422	IEC 60247	at 90 °C and 40-60 Hz	≤ 0.015	-	≤ 0.01
Conductivity						
Water Content	IEEE C57.106-2015	ASTM D1533	ppm. (Mg/kg) maximum	≤ 20	≤ 10	
	IEC 60422	IEC 60814	ppm. (Mg/kg) maximum	< 20	< 10	
Moisture in Oil						
Moisture in Paper						
Color	IEEE C57.106-2015	ASTM D1500	Units maximum	≤ 1	≤ 1	≤ 0.5
	IEC 60422	ISO 2049	-	≤ 2		
Interfacial tension (IFT)	IEEE C57.106-2015	ASTM D971	mN/m	≥ 38		
	IEC 60422	ASTM D971, EN 14210	mN/m	≥ 35		
Neutralization number (Acidity)	IEEE C57.106-2015	ASTM D974	Mg KOH/g maximum	≤ 0.03		
	IEC 60422	IEC 62021-1 or IEC 62021	Mg KOH/g maximum	≤ 0.03		
Visual examination	IEEE C57.106-2015	ASTM D1524	-	Bright and Clear		
	IEC 60422	ISO 2049	-	Clear, free from sediment and suspended matter		
Oxidation inhibitor content	IEEE C57.106-2015	ASTM D2668	Type I oil, % maximum	≤ 0.08		
			Type II oil, % maximum	≤ 0.3		
Total dissolved gas	IEEE C57.106-2015	ASTM D3612	-	N/A	N/A	0.5 % or per manufacturer's requirements
Corrosive sulfer	IEEE C57.106-2015	ASTM D1275	-	Non corrosive		
	IEC 60422	IEC 62535	-	Non corrosive		
III. Continued use of in-service mineral oil						
Dielectric breakdown	IEEE C57.106-2015	ASTM D1816	1 mm. gap (Spherically capped), kV	≥ 23	≥ 28	≥ 30
			2 mm. gap (Spherically capped), kV	≥ 40	≥ 47	≥ 50
	IEC 60422	IEC 60156	2.5 mm. gap (Spherical or Spherically cappe	> 40	> 50	> 60
Dissipation factor (Power Factor)	IEEE C57.106-2015	ASTM D924	at 25 °C % maximum (%)	≤ 0.5		
			at 100 °C % maximum (%)	≤ 5		
Conductivity	IEC 61620	-	pS/m	Light use ≤ 1	,Middle use ≤ 5	,Heavily used > 5
Water Content	IEEE C57.106-2015	ASTM D1533	ppm (Mg/kg) maximum	≤ 35	≤ 25	≤ 20
Moisture in Oil	IEEE 62-1995		%Sataration			
Moisture in Paper	IEEE 62-1995		%M/dw			
Color	IEEE C57.152-2013	ASTM D1500	-	≤ 2.5		
Interfacial tension (IFT)	IEEE C57.106-2015	ASTM D971	mN/m	≥ 25	≥ 30	≥ 32
Neutralization number (Acidity)	IEEE C57.106-2015	ASTM D974	mg KOH/g maximum	≤ 0.2	≤ 0.15	≤ 0.1
Corrosive sulfur		ASTM D1275	-			
Passivator	IEC 60666		ppm.	≥ 50		
Sludge	IEC60422	-	%w	No sediment or precipitable sludge. Results below	0.02	%w
Furan (2-FAL)		ASTM D5837	ppb. (Open Type)	< 1000		
			ppb. (Seal Type)	A < 100	100 > Q ≤ 250	U > 250
DP	Chendong	-	-	A > 700	450 < Q ≤ 700	U < 450
Oxidation inhibitor content	IEEE C57.106-2016	ASTM D2669	Type II mineral oil	0.08 % minimum if in original oil		
IV. Reconditioned or reclaimed mineral oil in transformers (After filling but before energizing)						
Dielectric breakdown	IEEE C57.106-2015	ASTM D1816	1 mm. gap (Spherically capped)	≥ 25	≥ 30	≥ 35
			2 mm. gap (Spherically capped)	≥ 45	≥ 55	≥ 60
	IEC 60422	IEC 60156	2.5 mm. gap (Spherical or Spherically capped)			
Water Content	IEEE C57.106-2015	ASTM D1533	ppm (Mg/kg) maximum	≤ 35	≤ 25	≤ 20
Interfacial tension (IFT)	IEEE C57.106-2015	ASTM D971	mN/m minimum	≥ 35		
Color	IEEE C57.106-2015	ASTM D1500	Maximum	≤ 1.5		
Visual examination	IEEE C57.106-2015	ASTM D1524	-	Clear		
Neutralization number (Acidity)	IEEE C57.106-2015	ASTM D974	mg KOH/g maximum	≤ 0.05		

Main Tank Transformer Oil Criteria (Mineral Oil)

Test Item	Standard	Method		Main Tank								
			Std. IEEE C57.106-2015	≤ 69 kV			> 69 kV - < 230 kV			≥ 230 kV		
			Std. IEC 60422	≤ 72.5 kV			> 72.5 kV - ≤ 170 kV			> 170 kV - < 400 kV		
I. New mineral oil (Received from the oil supplier)				Normal	Monitoring	Warning	Normal	Monitoring	Warning	Normal	Monitoring	Warning
Dielectric breakdown	IEEE C57.106-2015	ASTM D1816	1 mm. gap (Spherically capped), kV									
			2 mm. gap (Spherically capped), kV									
		ASTM D877	2.54 mm. or 0.1 in. gap (Disks), kV									
Dissipation factor (Power Factor)	IEEE C57.106-2015	ASTM D924	at 25 °C, % maximum									
			at 100 °C, % maximum									
Conductivity												
Water Content	IEEE C57.106-2015	ASTM D1533	ppm (Mg/kg) maximum									
Color	IEEE C57.106-2015	ASTM D1500	Units maximum									
Interfacial tension (IFT)	IEEE C57.106-2015	ASTM D971	mN/m									
Neutralization number (Acidity	IEEE C57.106-2015	ASTM D974	Mg KOH/g maximum									
Oxidation inhibitor content	IEEE C57.106-2015	ASTM D2668	Type I oil, % maximum									
			Type II oil, % maximum									
Corrosive sulfer	IEEE C57.106-2015	ASTM D1275	-									
Relative density (Specific Grav	IEEE C57.106-2015	ASTM D1298	at 15 °C maximum									
Visual examination	IEEE C57.106-2015	ASTM D1524	-									
II. New mineral oil (Received in new equipment or after filling,prior to energization)												
Dielectric breakdown	IEEE C57.106-2015	ASTM D1816	1 mm. gap (Spherically capped, kV)									
			2 mm. gap (Spherically capped, kV)									
	IEC 60422	IEC 60156	2.5 mm. gap (Spherical or Spherically capped, kV)									
Dissipation factor (Power Factor)	IEEE C57.106-2015	ASTM D924	at 25 °C, % maximum									
			at 100 °C, % maximum									
	IEC 60422	IEC 60247	at 90 °C and 40-60 Hz									
Conductivity												
Water Content	IEEE C57.106-2015	ASTM D1533	ppm. (Mg/kg) maximum									
	IEC 60422	IEC 60814	ppm. (Mg/kg) maximum									
Moisture in Oil												
Moisture in Paper												
Color	IEEE C57.106-2015	ASTM D1500	Units maximum									
	IEC 60422	ISO 2049	-									
Interfacial tension (IFT)	IEEE C57.106-2015	ASTM D971	mN/m									
	IEC 60422	ASTM D971, EN 14210	mN/m									
Neutralization number (Acidity	IEEE C57.106-2015	ASTM D974	Mg KOH/g maximum									
	IEC 60422	IEC 62021-1 or IEC 62021-2	Mg KOH/g maximum									
Visual examination	IEEE C57.106-2015	ASTM D1524	-									
	IEC 60422	ISO 2049	-									
Oxidation inhibitor content	IEEE C57.106-2015	ASTM D2668	Type I oil, % maximum									
			Type II oil, % maximum									
Total dissolved gas	IEEE C57.106-2015	ASTM D3612	-									
Corrosive sulfer	IEEE C57.106-2015	ASTM D1275	-									
	IEC 60422	IEC 62535	-									
III. Continued use of in-service mineral oil												
Dielectric breakdown	IEEE C57.106-2015	ASTM D1816	1 mm. gap (Spherically capped), kV	28	28-23	23	33	33-28	28	35	35-30	30
			2 mm. gap (Spherically capped), kV	45	45-40	40	52	52-47	47	55	55-50	50
	IEC 60422	IEC 60156	2.5 mm. gap (Spherical or Spherically capped), kV	45	45-40	40	55	55-50	50	65	65-60	60
Dissipation factor (Power Factor)	IEEE C57.106-2015	ASTM D924	at 25 °C % maximum (%)	0.4	0.4-0.5	0.5	0.4	0.4-0.5	0.5	0.4	0.4-0.5	0.5
			at 100 °C % maximum (%)	4	4-5	5	4	4-5	5	4	4-5	5
Conductivity	IEC 61620	-	pS/m	4	4-5	5	4	4-5	5	4	4-5	5
Water Content	IEEE C57.106-2015	ASTM D1533	ppm (Mg/kg) maximum	30	30-35	35	20	20-25	25	15	15-20	20
Moisture in Oil	IEEE 62-1995		%Sataration		-			-			-	
Moisture in Paper	IEEE 62-1995		%M/dw		-			-			-	
Color	IEEE C57.152-2013	ASTM D1500	-	2	2-2.5	2.5	2	2-2.5	2.5	2	2-2.5	2.5
Interfacial tension (IFT)	IEEE C57.106-2015	ASTM D971	mN/m	28	28-25	25	33	33-30	30	35	35-32	32
Neutralization number (Acidity)	IEEE C57.106-2015	ASTM D974	mg KOH/g maximum	0.17	0.17-0.2	0.2	0.12	0.12-0.15	0.15	0.07	0.07-0.1	0.1
Corrosive sulfur		ASTM D1275	-	3a	3a-4a	4a	3a	3a-4a	4a	3a	3a-4a	4a
Passivator	IEC 60666		ppm.	70	70-50	50	70	70-50	50	70	70-50	50
Sludge	IEC60422	-	%w	0.018	0.018-0.02	0.02	0.018	0.018-0.02	0.02	0.018	0.018-0.02	0.02
Furan (2-FAL)		ASTM D5837	ppb. (Open Type)	700	700-1000	1000	700	700-1000	1000	700	700-1000	1000
			ppb. (Seal Type)	100	100-250	250	100	100-250	250	100	100-250	250
DP	Chendong	-	-	700	700-450	450	700	700-450	450	700	700-450	450
Oxidation inhibitor content	IEEE C57.106-2016	ASTM D2669	Type II mineral oil	0.1	0.1-0.08	0.08	0.1	0.1-0.08	0.08	0.1	0.1-0.08	0.08
IV. Reconditioned or reclaimed mineral oil in transformers (After filling but before energizing)												
Dielectric breakdown	IEEE C57.106-2015	ASTM D1816	1 mm. gap (Spherically capped)	30	30-25	25	35	35-30	30	40	40-35	35
			2 mm. gap (Spherically capped)	50	50-45	45	60	60-55	55	65	65-60	60
	IEC 60422	IEC 60156	2.5 mm. gap (Spherical or Spherically capped)	45	45-40	40	55	55-50	50	65	65-60	60
Water Content	IEEE C57.106-2015	ASTM D1533	ppm (Mg/kg) maximum	30	30-35	35	20	20-25	25	15	15-20	20
Interfacial tension (IFT)	IEEE C57.106-2015	ASTM D971	mN/m minimum	40	40-35	35	40	40-35	35	40	40-35	35
Color	IEEE C57.106-2015	ASTM D1500	Maximum	1	1-1.5	1.5	1	1-1.5	1.5	1	1-1.5	1.5
Visual examination	IEEE C57.106-2015	ASTM D1524	-	Clear								
Neutralization number (Acidity)	IEEE C57.106-2015	ASTM D974	mg KOH/g maximum	0.03	0.03-0.05	0.05	0.03	0.03-0.05	0.05	0.03	0.03-0.05	0.05